Ex.No: 12

OBSERVATION:(Simulation)



Fig 1: *Selecting the clustering algorithm (Agglomerative Clustering)*

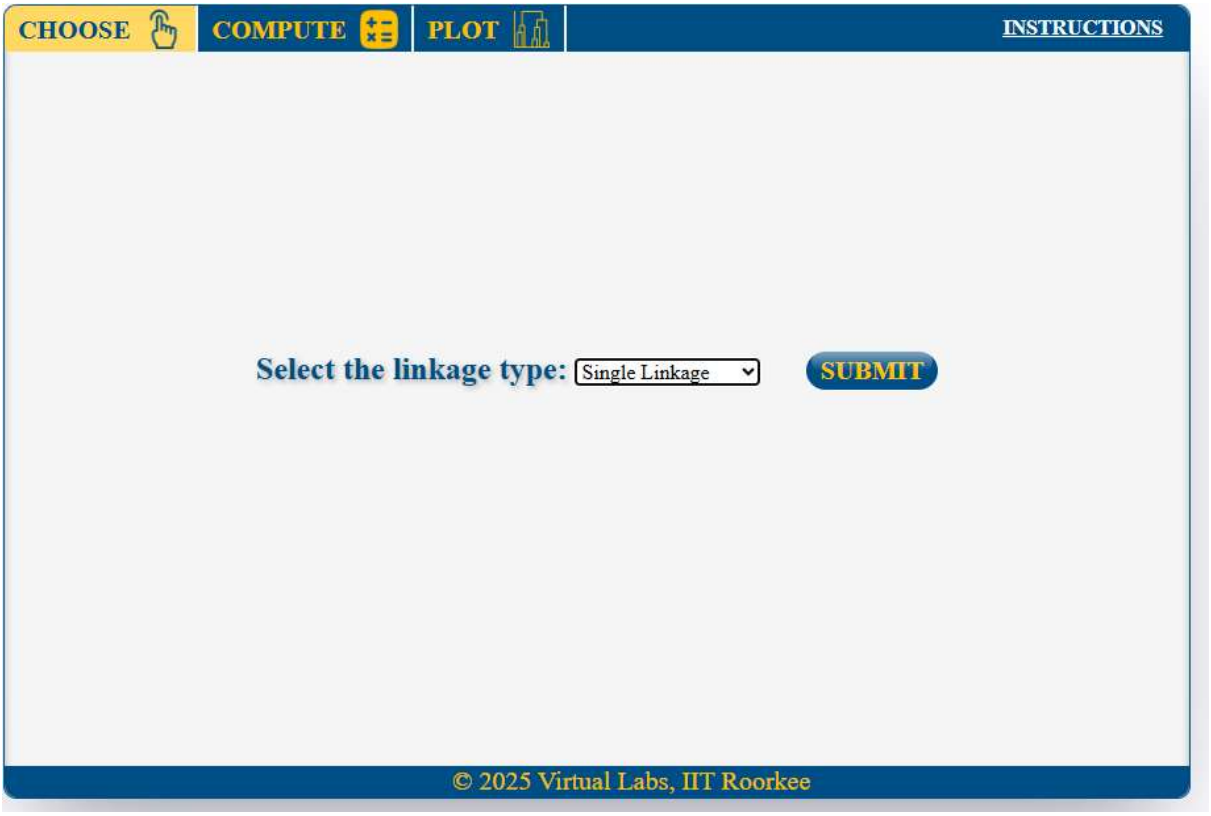


Fig 2: *Selecting the linkage type (Single Linkage)*

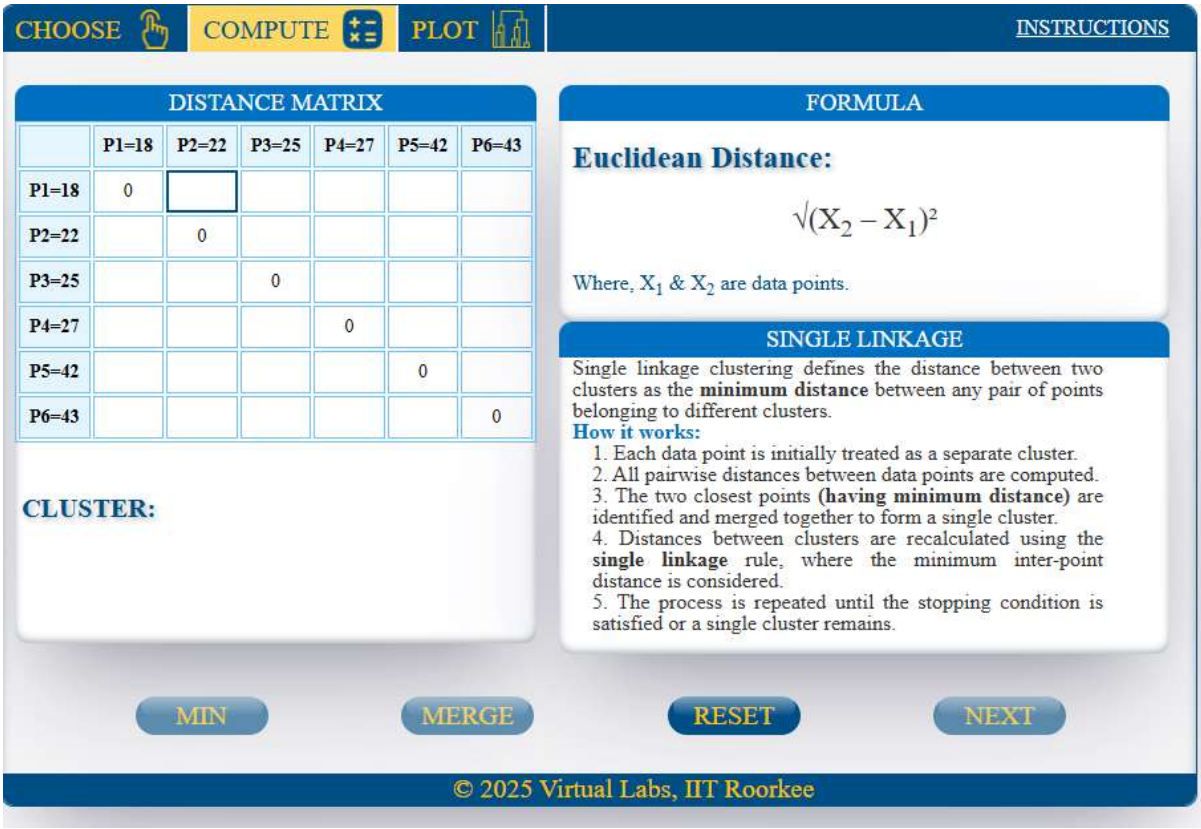


Fig 3: Initial distance matrix and formula display

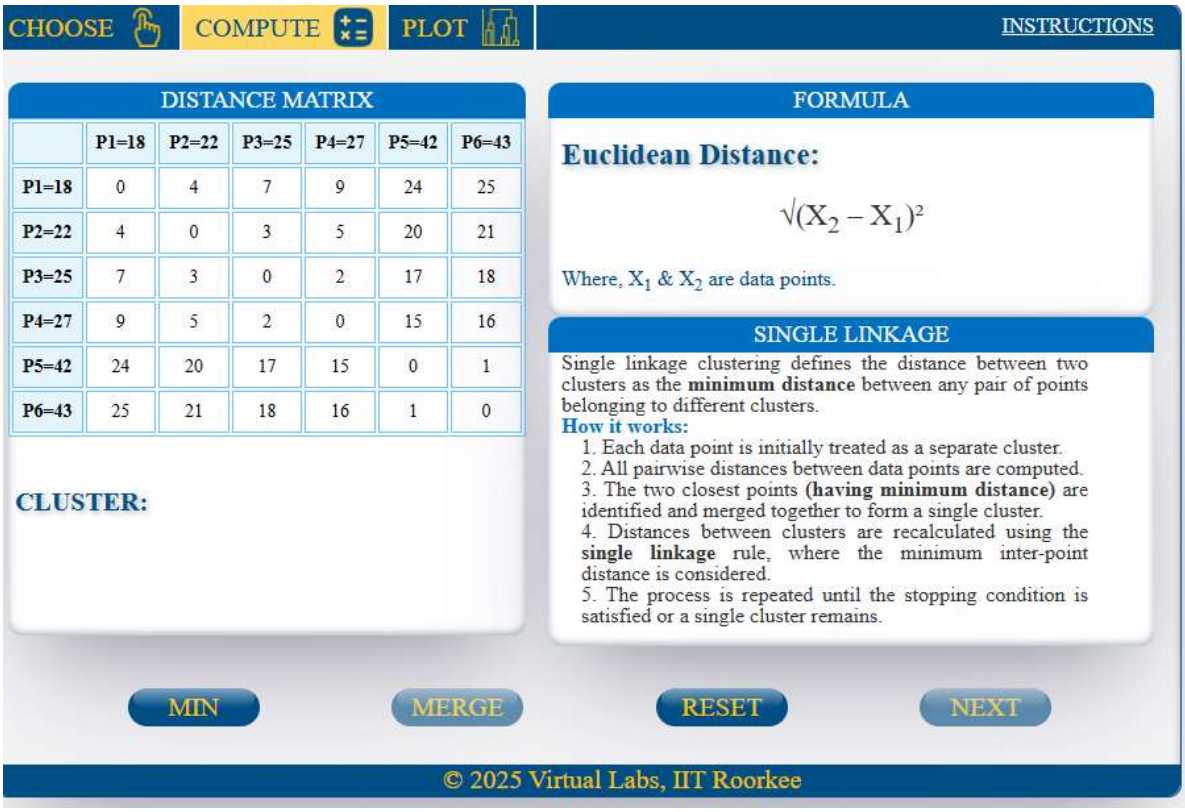


Fig 4: Calculated distance matrix for all data points

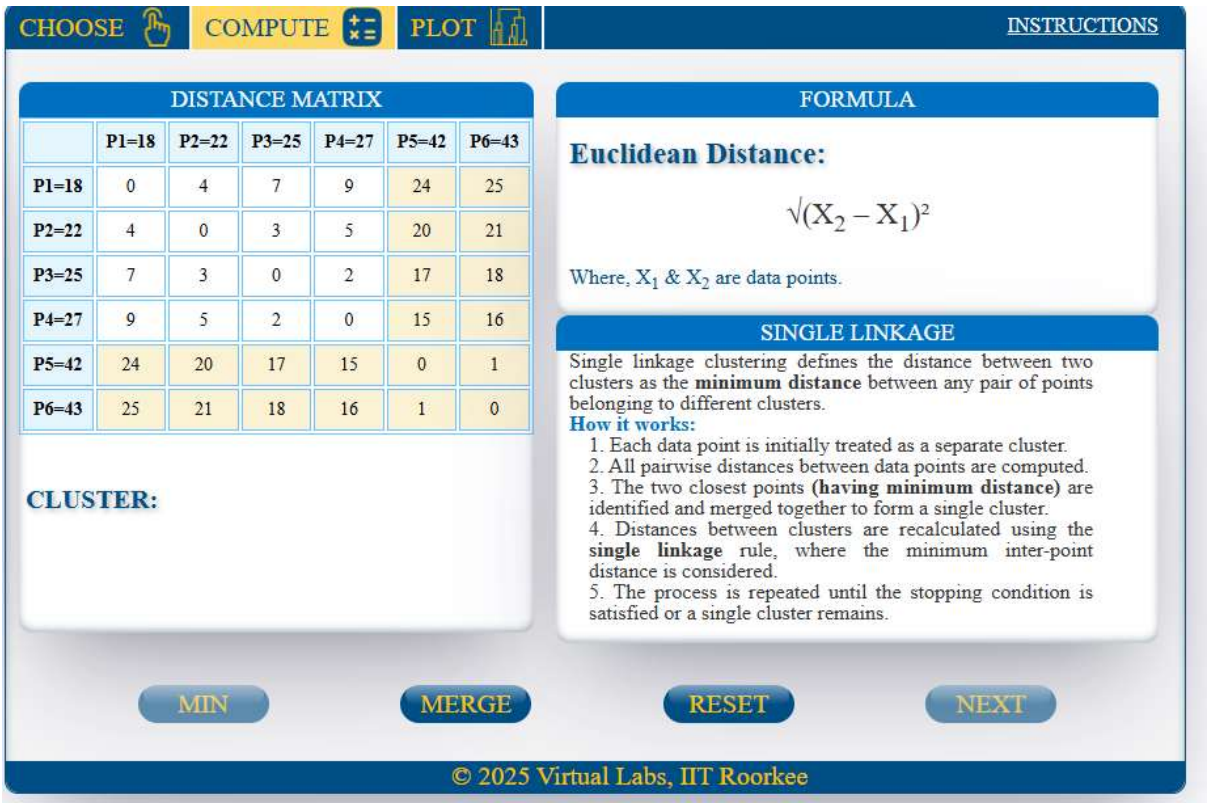


Fig 5: *Finding the minimum distance in the distance matrix*

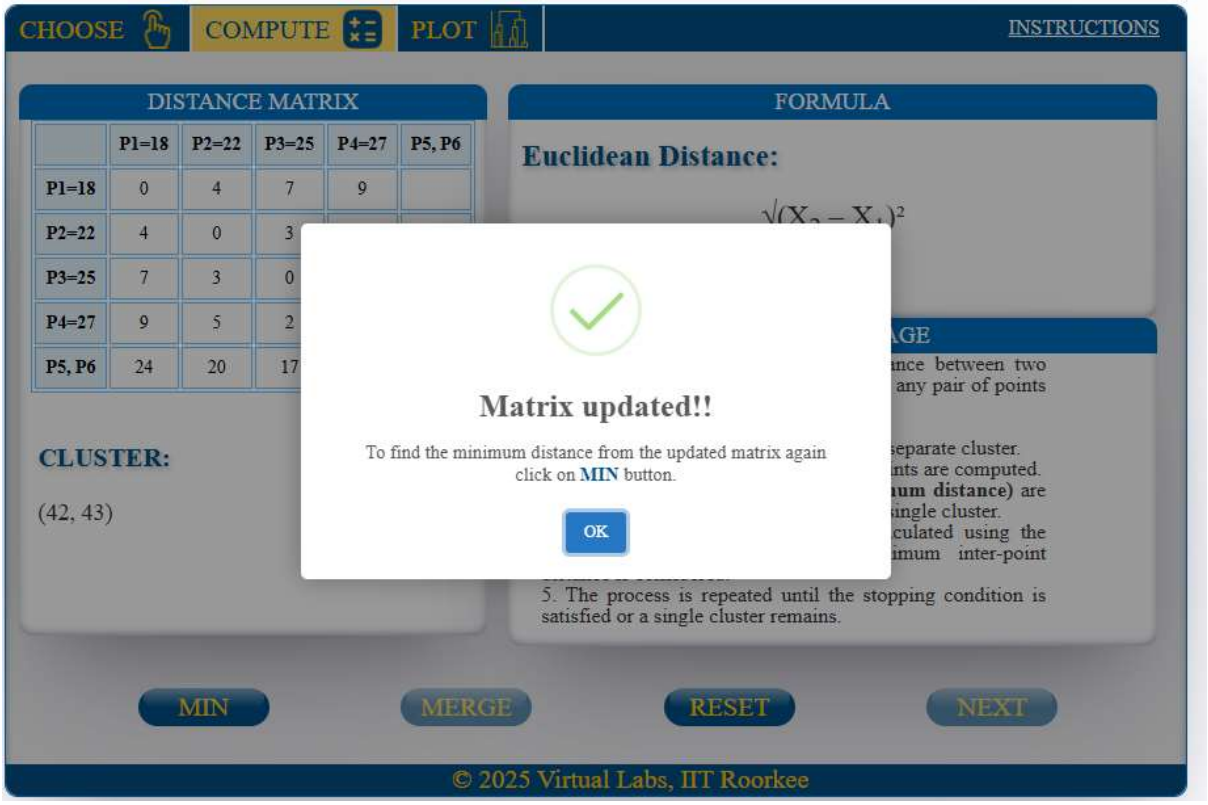


Fig 6: *Popup confirmation after updating the distance matrix for cluster(42, 43)*

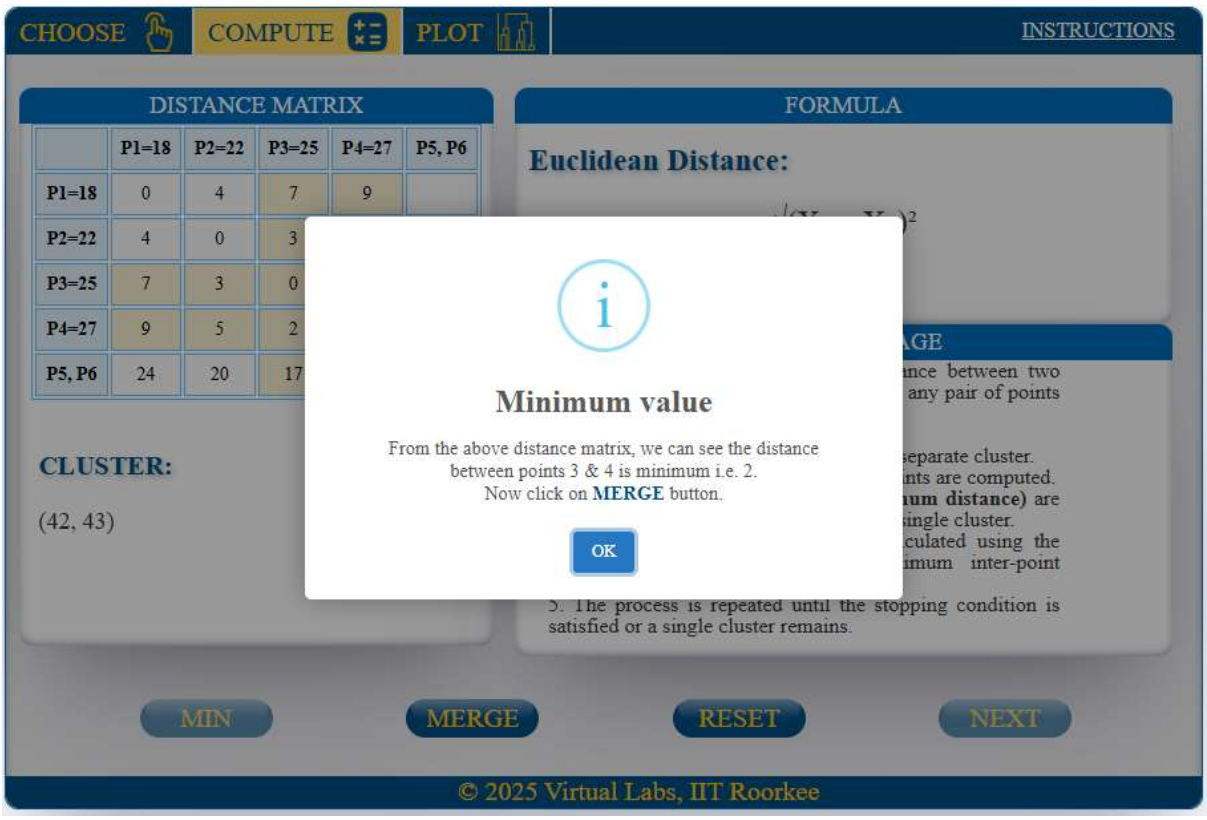


Fig 7: Highlighting minimum distance between points 3 and 4

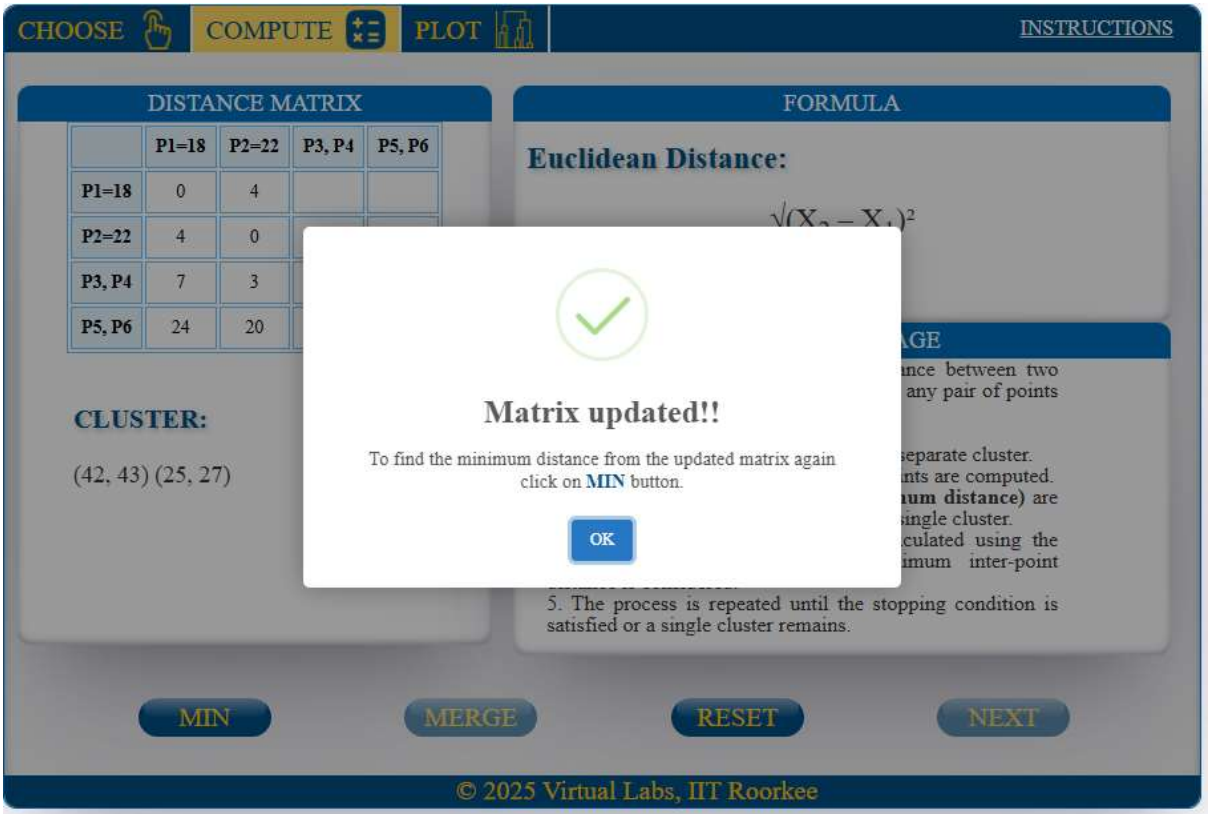


Fig 8: Popup confirmation after updating matrix for cluster (25, 27)

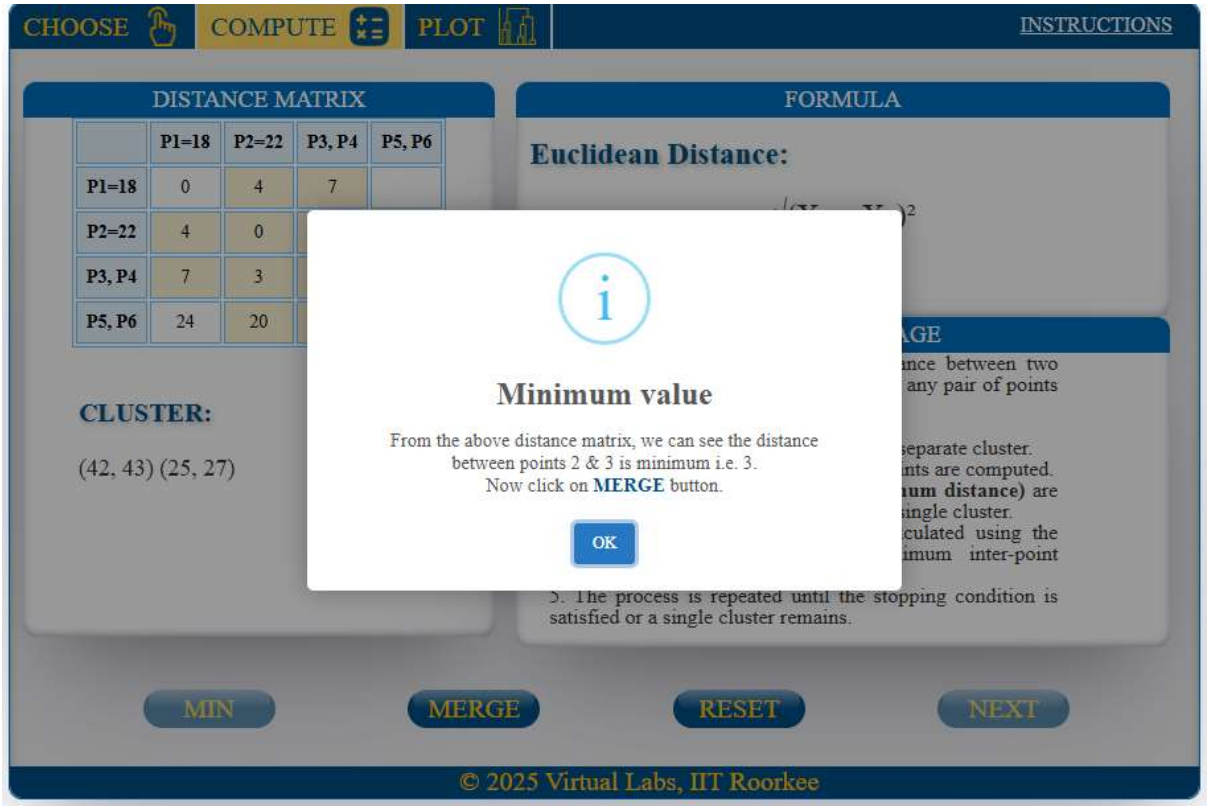


Fig 9: Highlighting minimum distance between point 2 and cluster (25, 27)

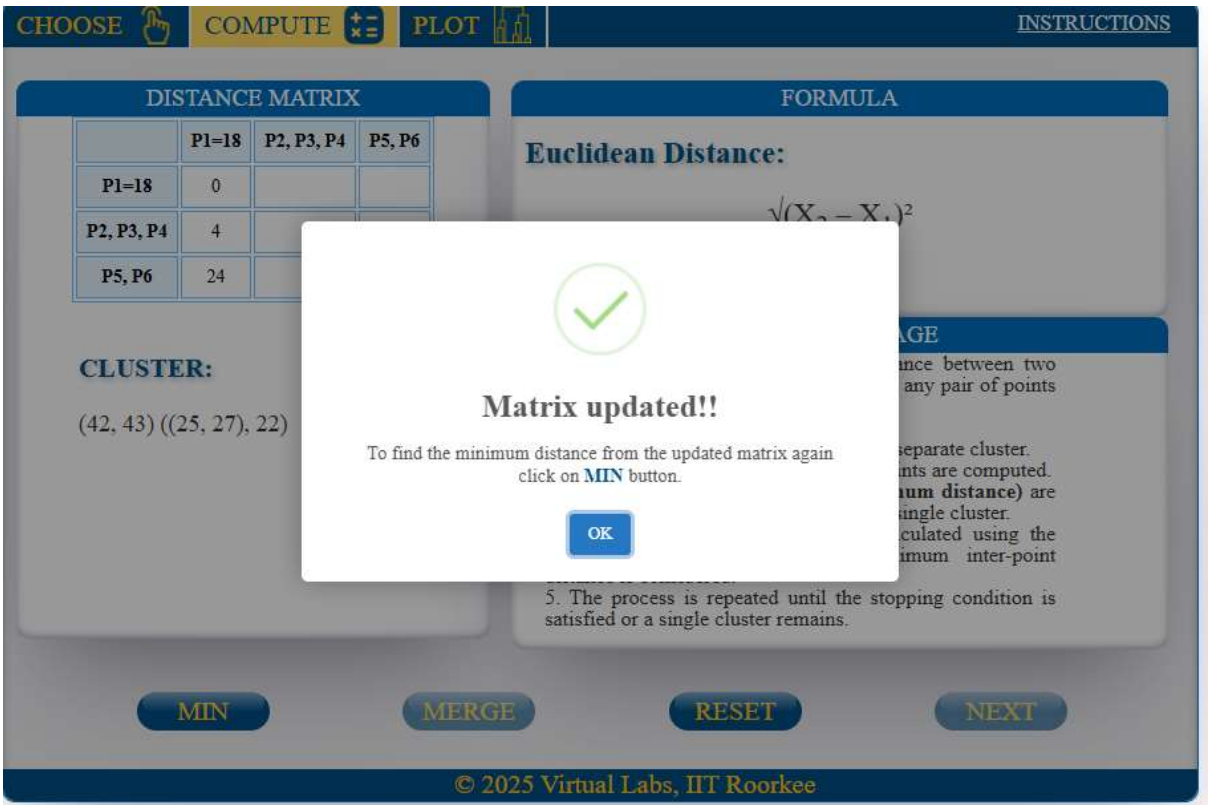


Fig 10: Popup confirmation after merging cluster ((25, 27), 22)

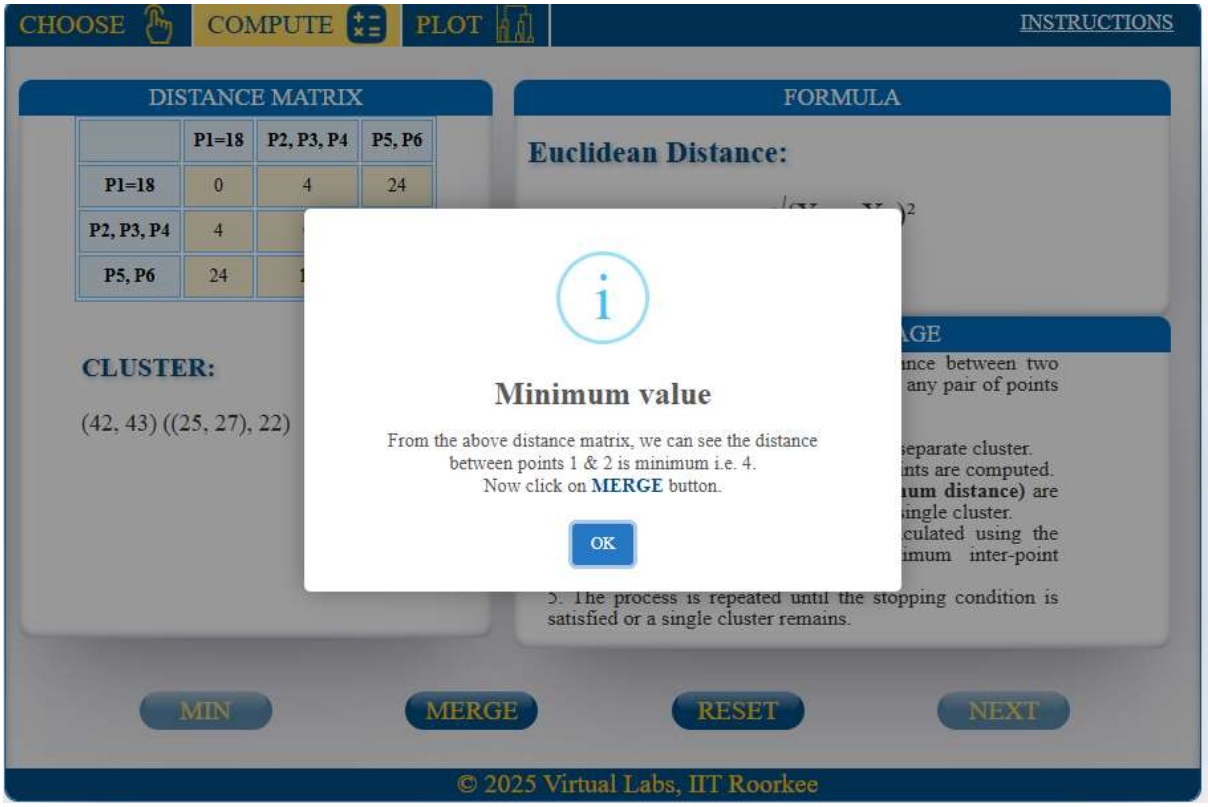


Fig 11: Highlighting minimum distance between point 1 and cluster((25, 27),22)

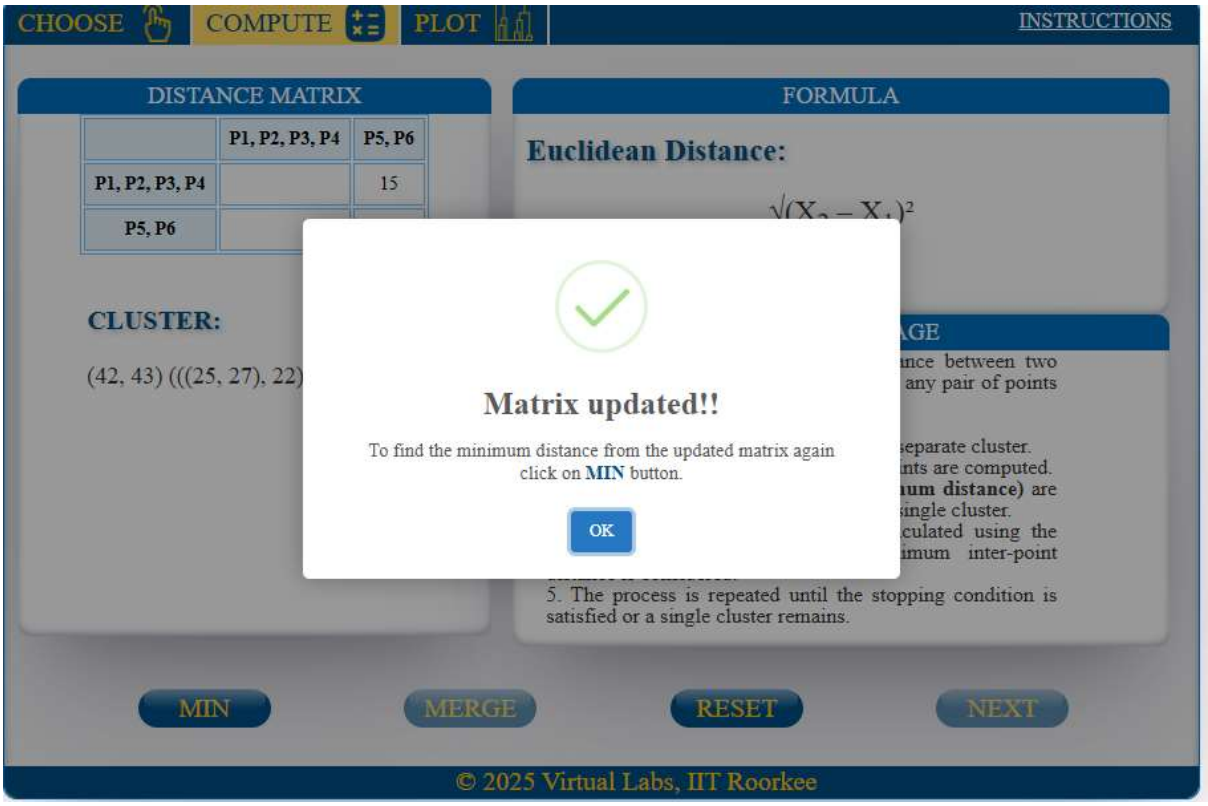


Fig 12: Popup confirmation after updating matrix for cluster(((25, 27), 22), 18)

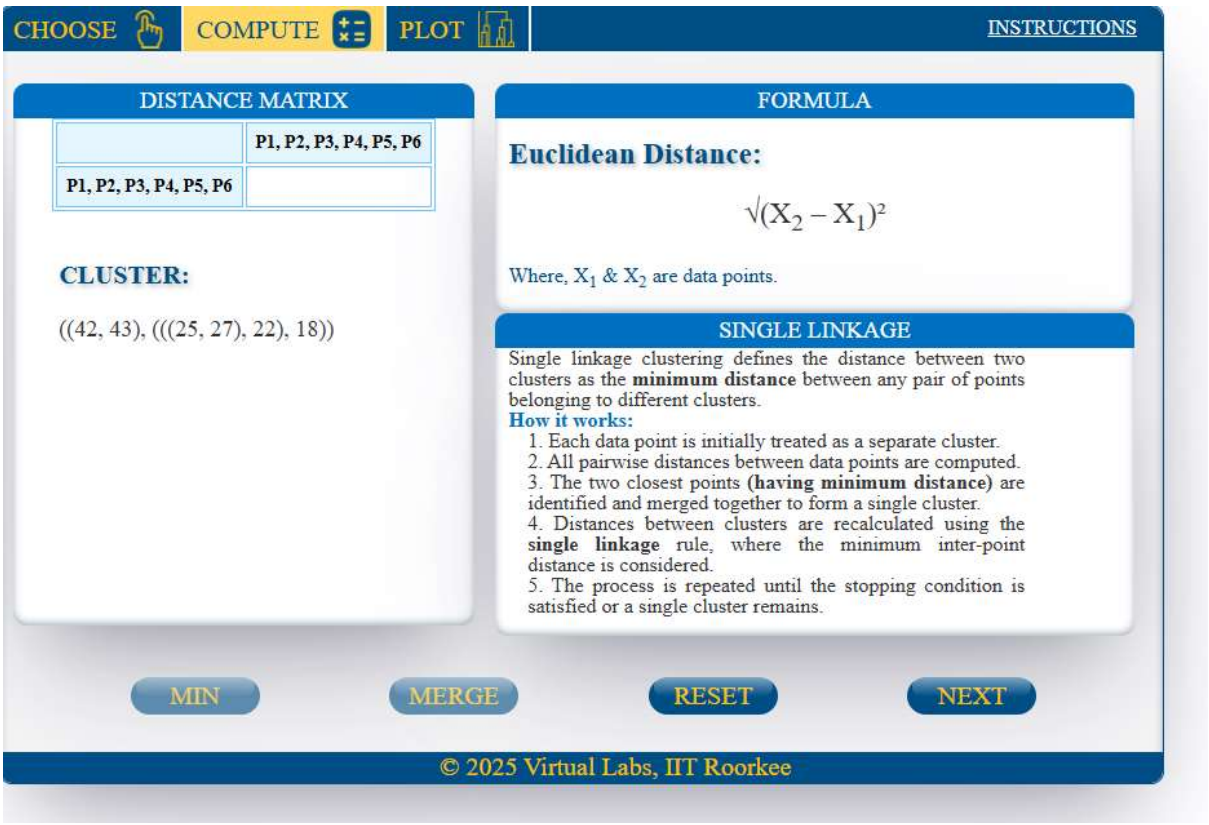


Fig 13: *Final distance matrix calculation before total cluster merge*



Fig 14: *Prompt message asking to plot the dendrogram*



Fig 15: Viewing selected data points list

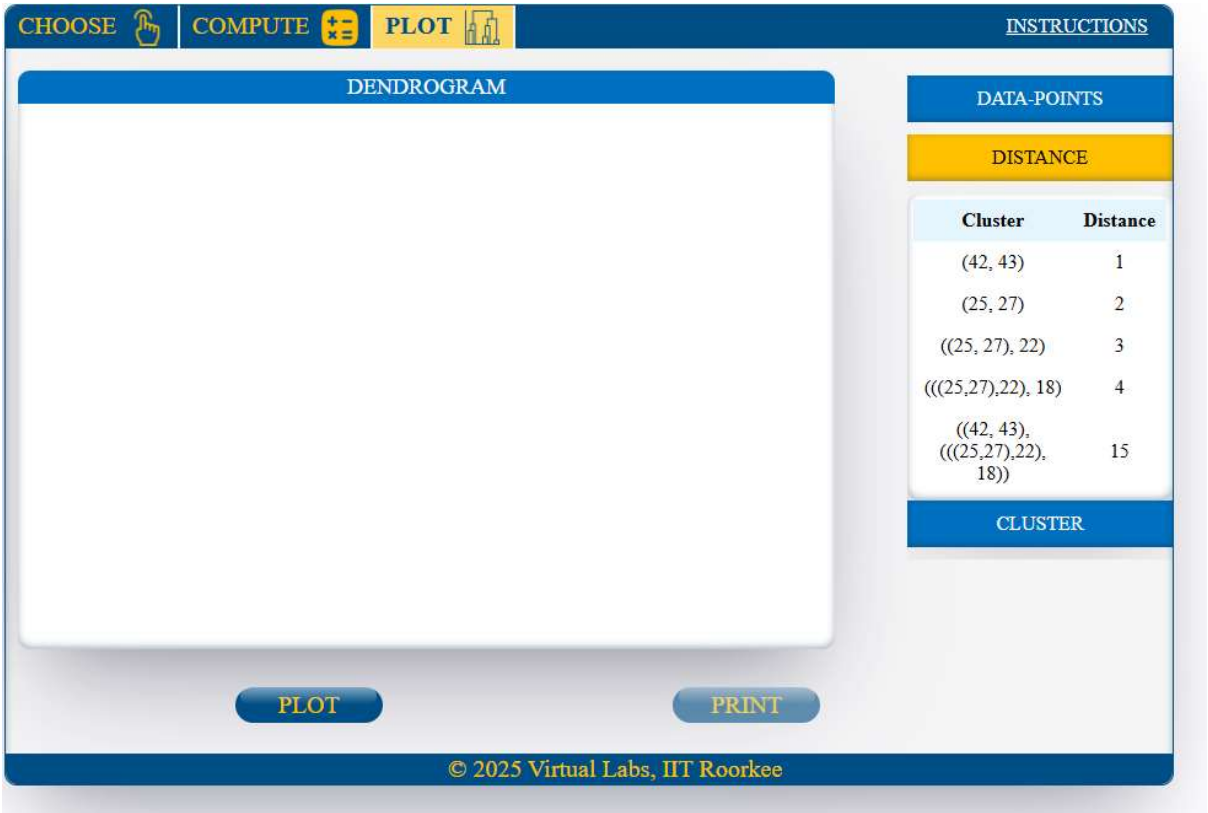


Fig 16: Viewing cluster distance values table



Fig 17: *Viewing final merged cluster output string*

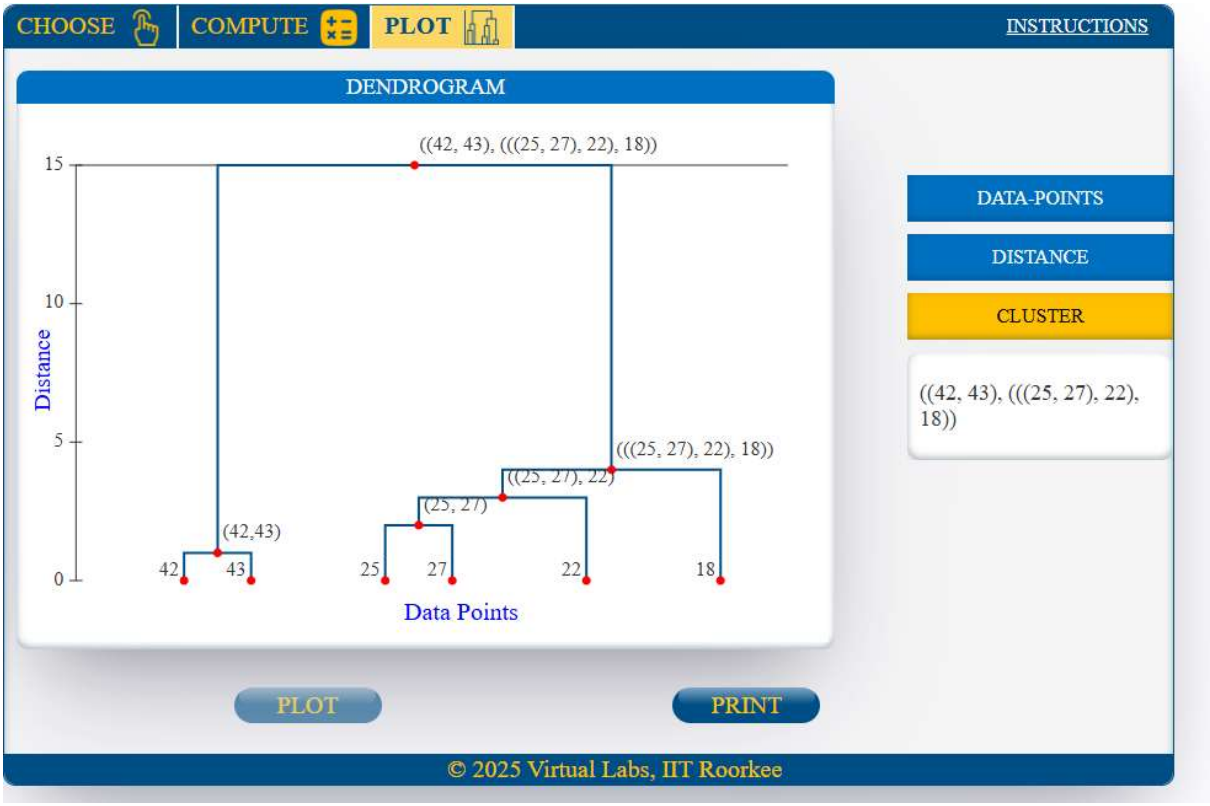


Fig 18: *Final plotted dendrogram showing full hierarchical cluster tree*

Simulation link: https://ml-iitr.vlabs.ac.in/exp/hierarchical-clustering/simulation.html